

AI-POWERD CONGNITIVE SCREENING AND ANALYSIS PLATFORM

25-26J-439



CogniCare: An AI-Powered Cognitive Screening & Analysis Platform

Problem Background

- Dementia: 55M+ cases worldwide, 10M new annually (WHO 2021).
- Sri Lanka: rising elderly population, limited affordable screening tools.
- **Early detection is critical** to slow progression and improve quality of life.
- Regular hospital visits are **inconvenient** due to transport issues and time burden.



CogniCare: An AI-Powered Cognitive Screening & Analysis Platform

Impact:

Early detection enables timely treatment and support, improving the patient's quality of life while easing the emotional and practical burden on caregivers.

Research Gaps:

There is **no unified, explainable, culturally localized, and multimodal AI system** that integrates handwriting, MRI, facial/behavioral, and wearable data to provide both **early dementia screening (Demented/Non-Demented)** and **stage classification** in a **practical, affordable, and elderly-accessible platform** for Sri Lanka and similar low-resource contexts.



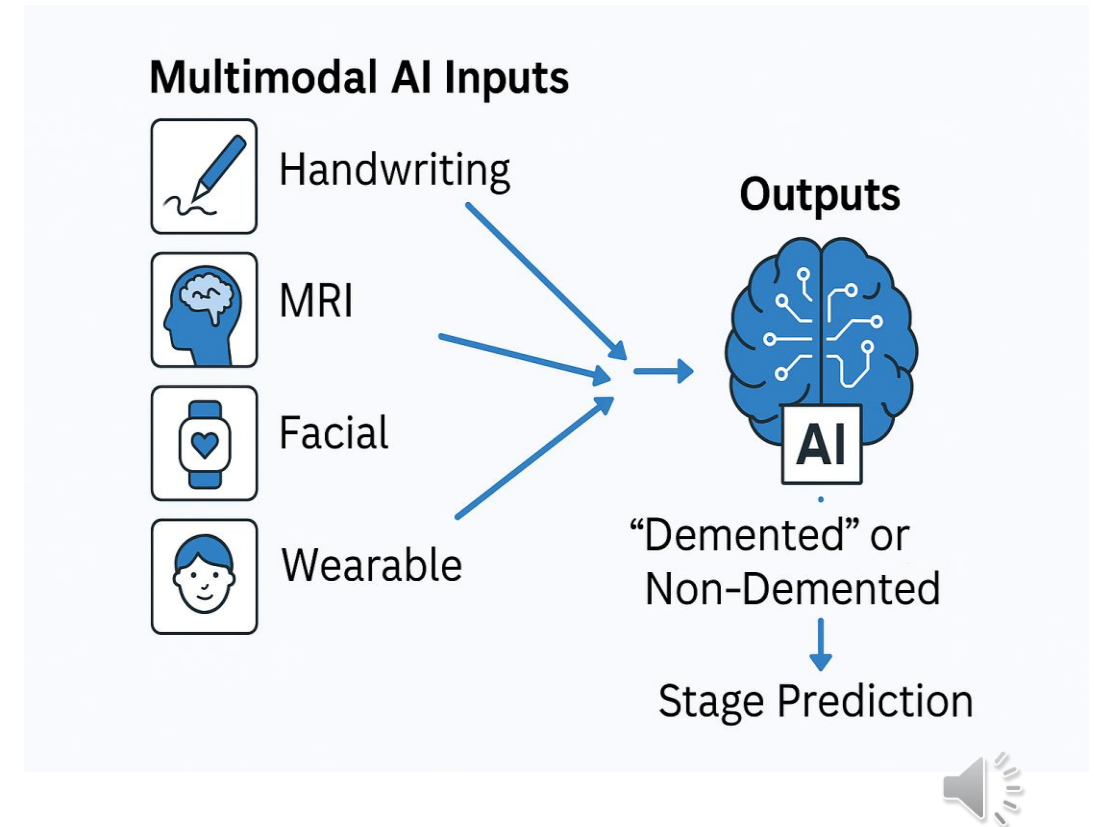
CogniCare: An AI-Powered Cognitive Screening & Analysis Platform

Proposed Solution

- **CogniCare Platform:** Mobile & web-based screening and analysis.
- **Multimodal AI Inputs:**
 - Handwriting stroke analysis (speed, pressure, jerk).
 - MRI-based dementia stage classification.
 - Facial expression & behavior analysis.
 - Wearable data (heart rate, sleep, activity, temperature).

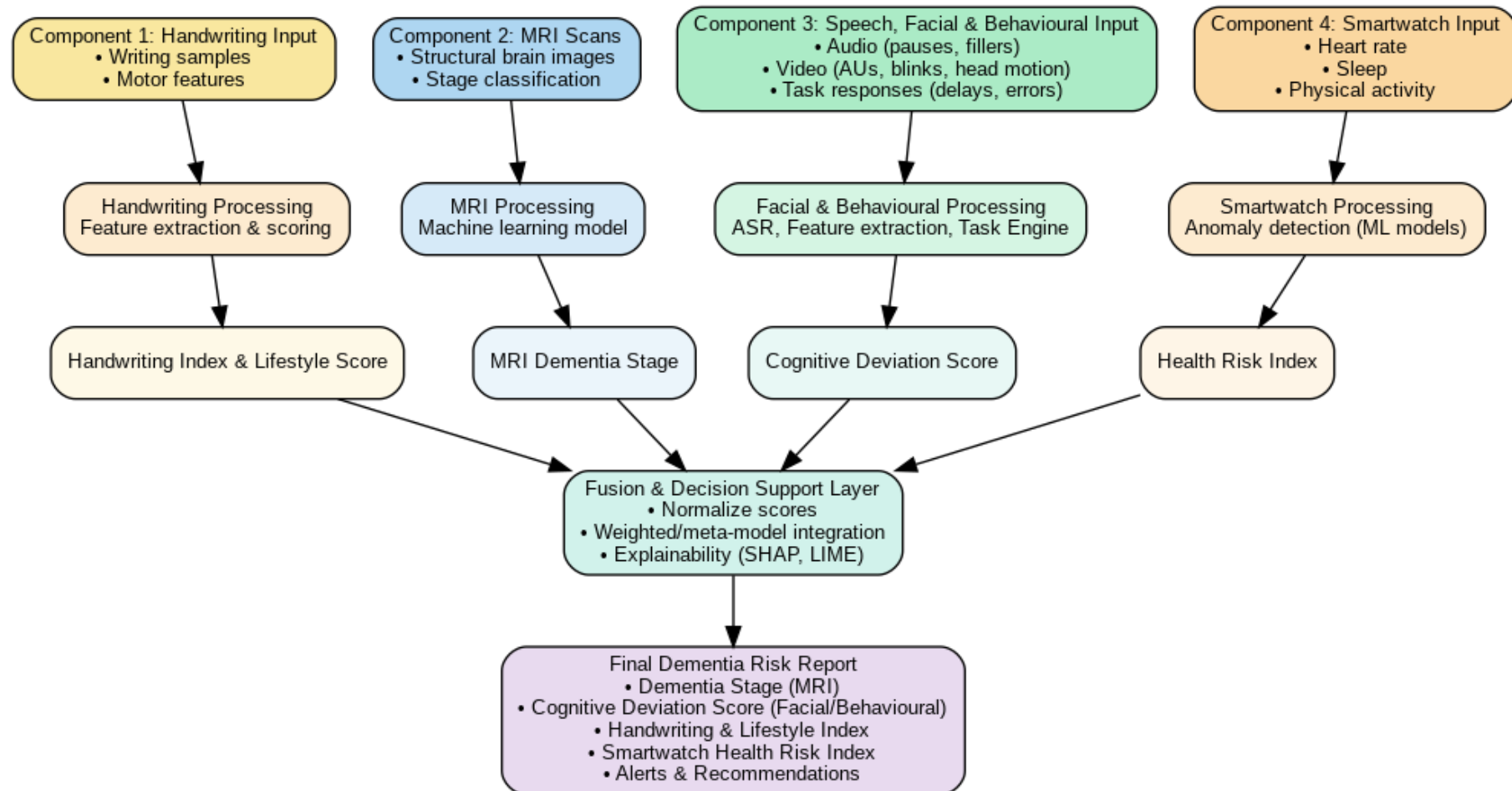
Outputs:

- Detects whether the patient is **Demented or Non-Demented**.
- For patients classified as Demented → predicts the **stage**



CogniCare: An AI-Powered Cognitive Screening & Analysis Platform

System Workflow



CogniCare: An AI-Powered Cognitive Screening & Analysis Platform

- **Constraints & Limitations:**
 - Data privacy & ethical concerns.
 - Limited clinical datasets.
 - Usability for elderly in low-resource settings.





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Handwriting-Based Dementia Screening Tool for Mobile and Web Platforms

Software Engineering

Handwriting-Based Dementia Screening Tool for Mobile and Web Platforms

Knowledge Gap:

- Most handwriting datasets are Western-centric.
- Costly equipment dependence (**Wacom tablets**)
- Limited explainability in ML models

Shortcomings of Existing Work:

- Dependence on costly devices like Wacom tablets
- Western-biased dataset
- Black-box AI with poor explainability.
- Not mobile-friendly

Proposed Solution:

- Mobile-ready, low-cost screening
- Augment existing datasets with local datasets.
- Explainable AI with SHAP for transparency.



Handwriting-Based Dementia Screening Tool for Mobile and Web Platforms

Domains of Knowledge Utilized:

AI& ML, Software Engineering, Data Science & Signal Processing (Handwriting feature extraction), Healthcare Informatics

Latest Technologies Used:

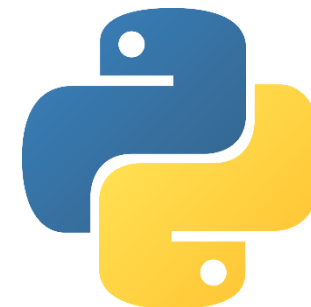
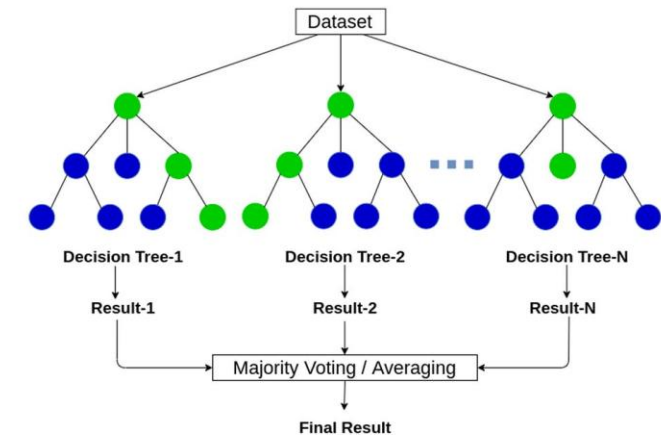
- TensorFlow Lite
- Scikit-learn & SHAP
- Firebase
- Docker

Novelty

- Culturally Adapted Handwriting & Multimodal Data
- Web & Mobile-Responsive Explainable AI



Random Forest



Handwriting-Based Dementia Screening Tool for Mobile and Web Platforms

Evaluation & Validation:

- **Datasets:** DARWIN (baseline reference) + local samples (local validation)
- **Metrics:** Accuracy, specificity, AUC (Area Under the Curve)
- **Validation methods:** Cross-validation, model explainability via SHAP.
- **Pilot testing**

Data Availability & Ethics:

- DARWIN dataset (publicly available) - measure the **motor control** of handwriting
- Augmented with locally collected data.
- Ethical clearance for clinical testing



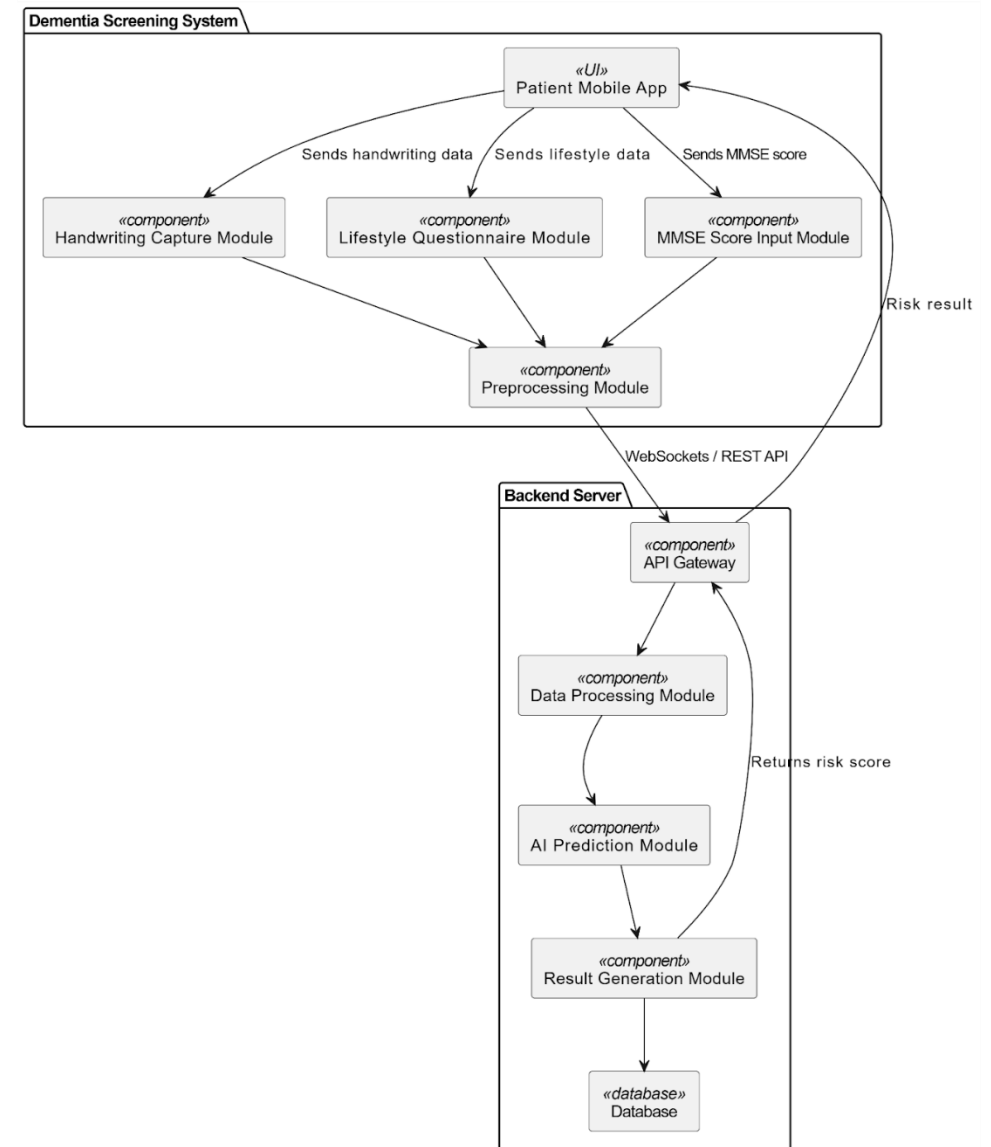
Handwriting-Based Dementia Screening Tool for Mobile and Web Platforms

System Overview of Components:

- **Data Collection Layer:** Handwriting input, lifestyle questionnaire
- **Feature Extraction Layer:** Kinematic features
- **Machine Learning Layer:** Model training + prediction
- **Output Layer:** Results via Mobile / Web app

Real-World Use Case: A household in Sri Lanka.

- Elderly patients use a tablet/phone at home for handwriting tasks + lifestyle questionnaire.
- AI model analyses inputs
- Output: Demented / Non-demented
- Explains to patient & caregiver.





IT22901644 | DISSANAYAKE M.P.S.

Dementia stage classification from MRI using Vision Transformer and explainable AI.
Software Engineering

Dementia stage classification from MRI using Vision Transformer and explainable AI.

Knowledge Gap:

- Existing methods mostly use **binary classification** (healthy vs diseased).
- Lack of **explainability**, making AI results hard to trust clinically.

Shortcomings of Existing Work:

- CNNs fail to capture long-range spatial dependencies in MRI.
- No visual reasoning for predictions.

Proposed Solution:

- Multi-stage dementia classification (CDR: 0, 0.5, 1, 2).
 - Non-Demented
 - Very Mild Dementia
 - Mild Dementia
 - Moderate Dementia
- Vision Transformer (ViT) + Grad-CAM for accuracy **and** clinical transparency.



Dementia stage classification from MRI using Vision Transformer and explainable AI.

Key Domains Used: Machine Learning, Deep Learning, Image Processing, Explainable AI.

Technologies:

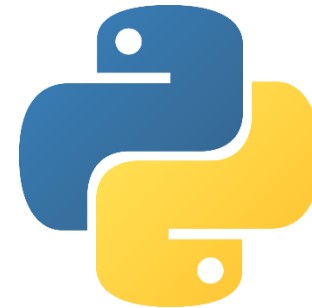
- Vision Transformers (global spatial feature extraction).
- Grad-CAM for heatmap visualizations.
- PyTorch and TensorFlow for model building.

Data & Ethics:

- OASIS dataset (public MRI scans with CDR labels).
- De-identified data to meet ethical standards.

Success Metrics:

- Accuracy $\geq 85\%$, plus precision, recall.
- Clinical validation through heatmap interpretability.



Dementia stage classification from MRI using Vision Transformer and explainable AI.

Novelty of Approach:

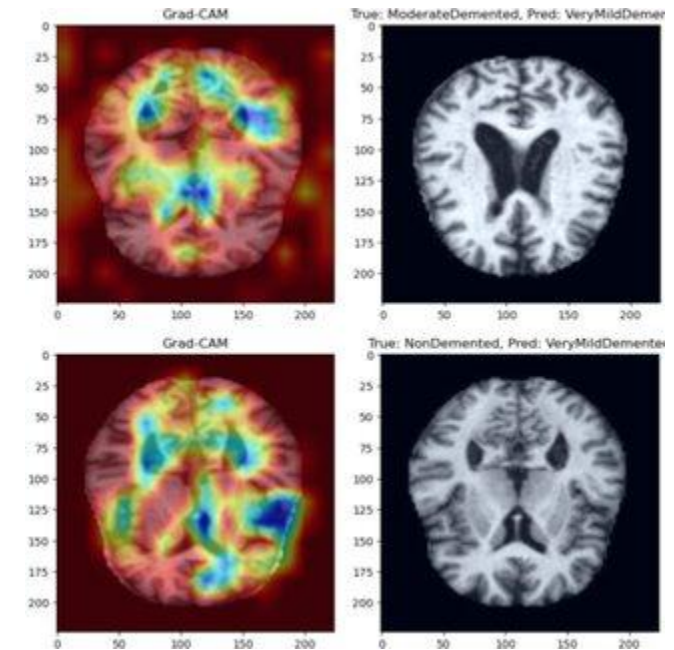
- Moves from binary to **fine-grained staging**, aligned with real clinical needs.
- Balances predictive power with interpretability.

How It Will Be Validated:

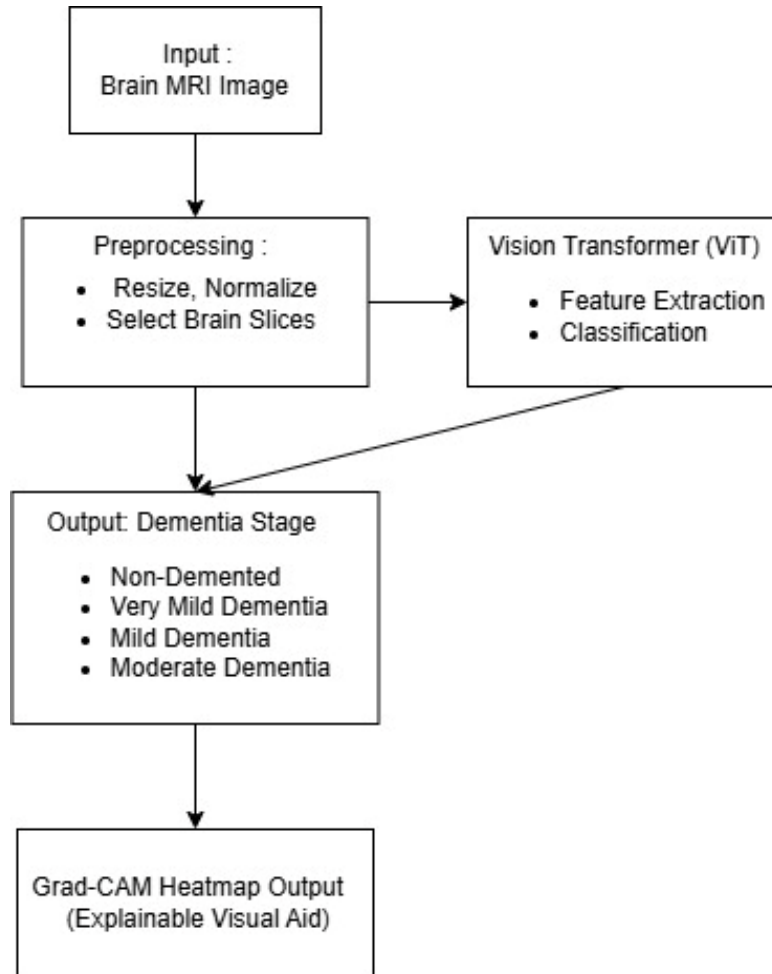
- Train/Validation/Test splits on OASIS dataset.
- Error analysis on misclassifications.

Advantages :

- Clear stage predictions with confidence scores.
- Heatmaps highlight brain regions influencing decisions.



Dementia stage classification from MRI using Vision Transformer and explainable AI.



System Overview:

- **Input:** T1-weighted MRI scan.
- **Pre-processing:** Resize, normalize, denoise.
- ViT model performs 4-class dementia staging.
- Grad-CAM generates heatmaps for transparency.
- **Output:** Predicted stage + probability + heatmap.

Real-World Usage:

- User uploads scan → gets stage prediction with visual evidence.
- Report feature to export classification results into patient records.



IT22358820 | PIRUTHIVI K

SMART HEALTH MONITORING AND DEMENTIA RISK ANALYSIS

Software engineering

SMART HEALTH MONITORING AND DEMENTIA RISK ANALYSIS

Knowledge Gap:

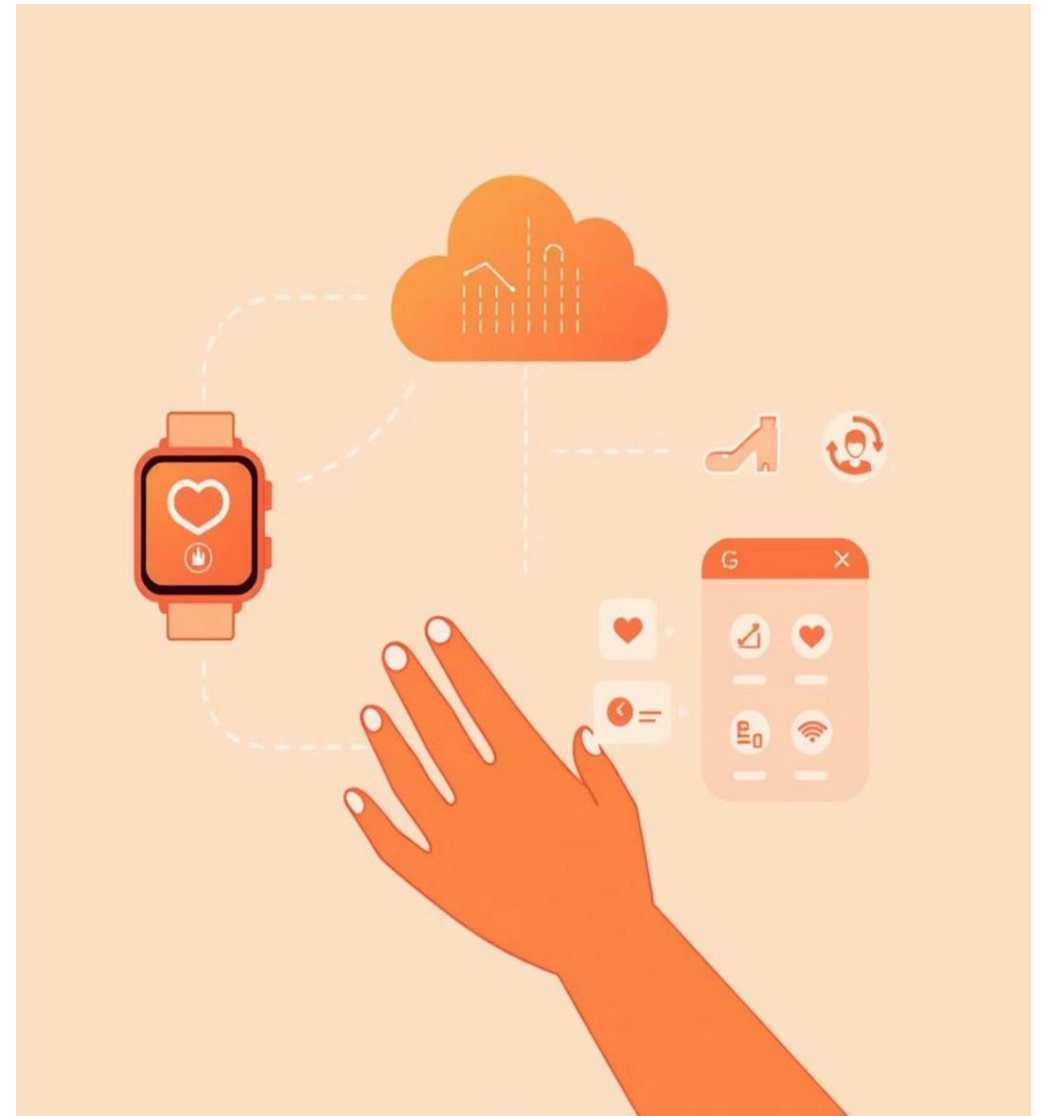
- Existing systems:
- ✓ Focus on monitoring only
- ✓ Depend on clinical → costly, less accessible
- ✓ No personalized AI-based lifestyle guidance
- ✓ Lack of cognitive tools

Shortcomings of Existing Work/novelty:

- Data monitoring without preventive action
- Alerts exist but no daily health nudges
- Underutilization of affordable wearables
- Solutions work in isolation

Proposed Solution:

- Risk scoring: Normal / Medium / High
- Personalized lifestyle tips → hydration, walking, stretching
- Cognitive & emotional engagement → eg: memory games
- Caregiver alerts for abnormalities

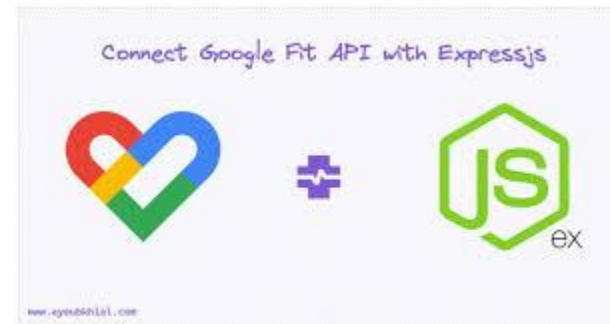


SMART HEALTH MONITORING AND DEMENTIA RISK ANALYSIS

Key Domains of Specialization-Based Knowledge

- **AI & ML:** Risk analysis, anomaly detection, predictive modeling .
- **Wearable Computing:** Real-time collection of physiological data via smartwatches.
- **UI/ UX Design:** Designing elderly-friendly mobile interfaces
- **Health Informatics:** clear Understanding dementia-related health parameters

Latest Technologies Utilized



SMART HEALTH MONITORING AND DEMENTIA RISK ANALYSIS

Evaluation & Validation

- Metrics: Accuracy, Recall, F1-score
- Testing: Simulated smartwatch data + real-world users
- Validation: Risk classification, lifestyle nudges, and alerts measured across both datasets and live wearable data accuracy and engagement compared to monitoring-only systems

Data Availability & Ethical Considerations

• Data Sources:

Public dementia datasets (Kaggle)

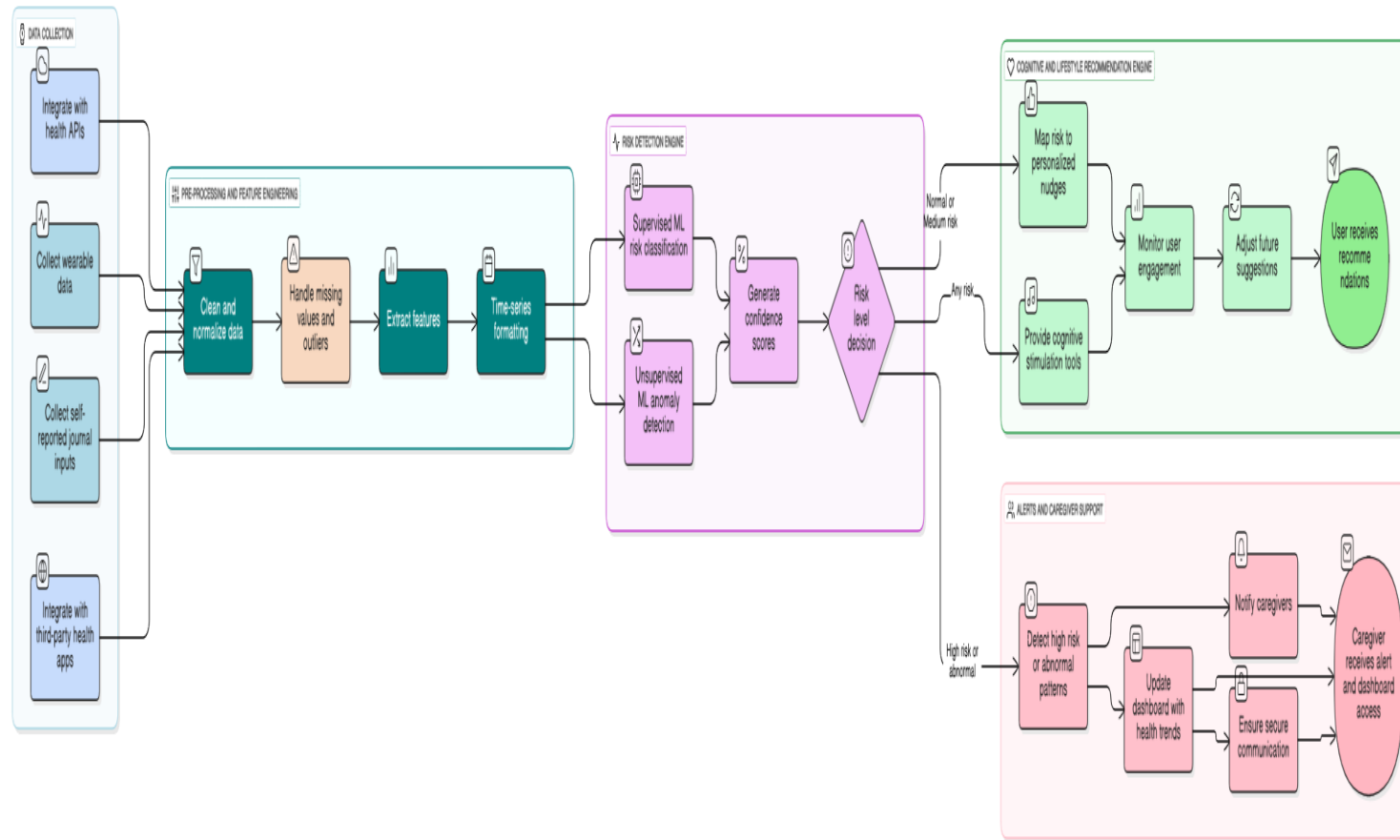
Real-time wearable data

Ethical Clearance:

Ensures Data anonymization & encryption to protect privacy

Compliance with local & international health data laws (GDPR/HIPAA if relevant)

SMART HEALTH MONITORING AND DEMENTIA RISK ANALYSIS



Real-World Application:

- user wears smartwatch → continuous monitoring
- AI flags risks → lifestyle tips + caregiver notifications
- Daily engagement: puzzles, memory recall, music therapy, storyboards
- Caregivers alerted in emergencies

Unique Impact:

- Beyond monitoring → provides action + engagement
- Unified web-first solution → dementia risk *classification + guidance + support* home-based Unified web-first solution for the dementia
- Enhances both physical & mental well-being



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DEMENTIA SCREENING VIA SPEECH, FACIAL EXPRESSIONS, AND TASK BEHAVIOUR

Software engineering

DEMENTIA SCREENING VIA SPEECH, FACIAL EXPRESSIONS, AND TASK BEHAVIOUR

Knowledge Gap

- Speech-only approaches: ignore facial/behavioural cues.
- Wearables: focus on physiology, not cognition.
- Few systems integrate multimodal cues dementia risk.

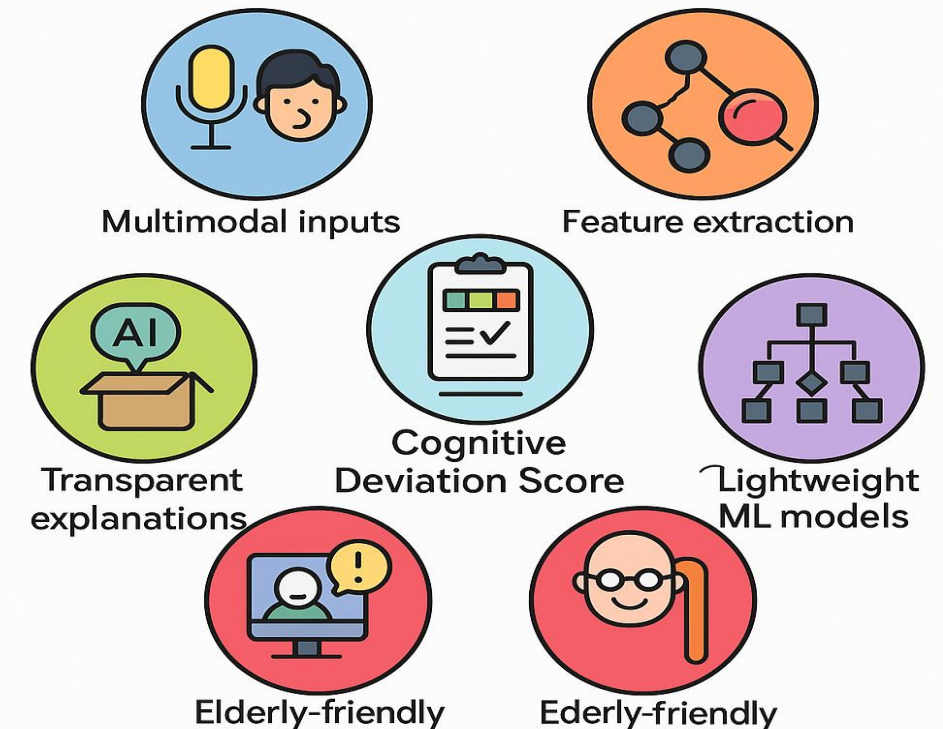
Shortcomings in Past Research

- Most studies use single modality (eg - speech).
- Limited focus on home-friendly, non-invasive systems.
- Lack of explainability in AI predictions.

Key Domains of Specialization-Based Knowledge

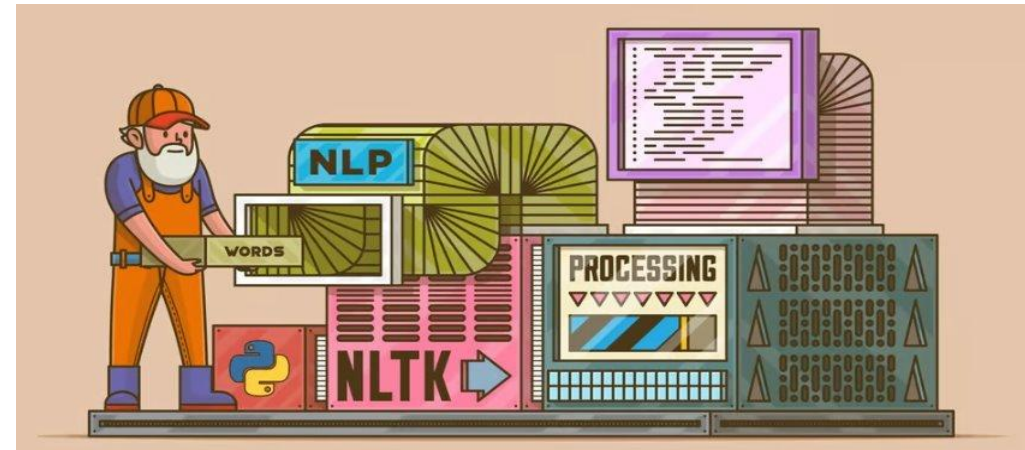
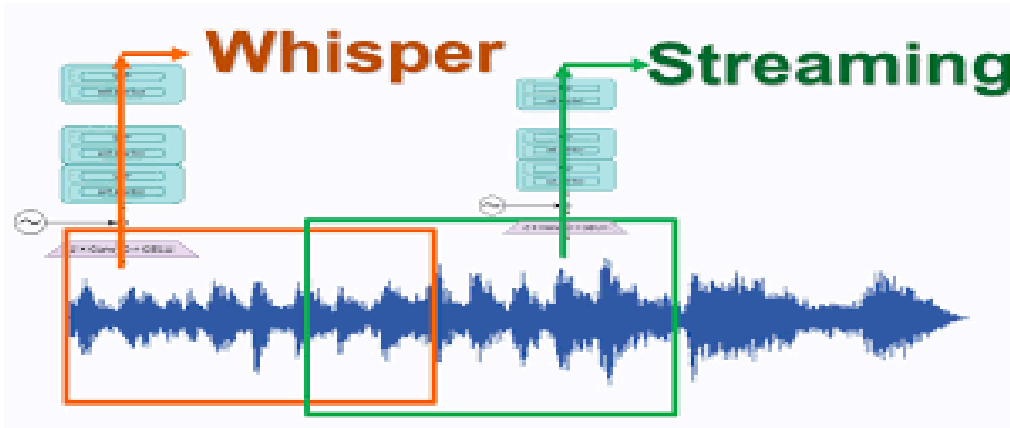
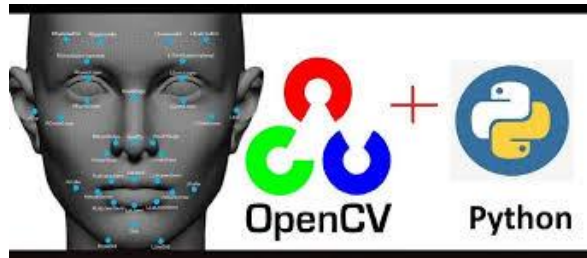
- **AI & Machine Learning:** Risk scoring, anomaly detection, explainable AI.
- **Computer Vision:** Facial AU analysis, blink/head motion tracking.
- **Speech Processing:** Transcription, pauses, fillers, word diversity.
- **HCI / UX Design:** Elderly-friendly screening interface.
- **Health Informatics:** Behavioural markers of dementia.

Proposed Solution



DEMENTIA SCREENING VIA SPEECH, FACIAL EXPRESSIONS, AND TASK BEHAVIOUR

Latest Technologies Utilized



- **ML Models:** Logistic Regression, Random Forest, Isolation Forest.

DEMENTIA SCREENING VIA SPEECH, FACIAL EXPRESSIONS, AND TASK BEHAVIOUR

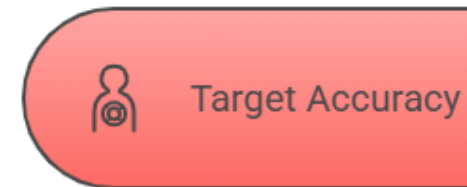
Data & Ethics:

- DementiaBank (speech) and ADReSS/ADReSSo datasets.
- PROMPT / OpenFace datasets for facial expressions (with access approval).
- All data de-identified to protect privacy.
- Explicit participant consent and ethical compliance.

Novelty of Approach:

- Integrates speech, facial expressions, and behavioural task data into a unified risk score.
- Moves beyond speech-only or MRI-only methods by capturing multimodal early markers.
- Uses explainable AI (SHAP, LIME) so clinicians and caregivers can see *why* the system flagged risk.

Success Metrics



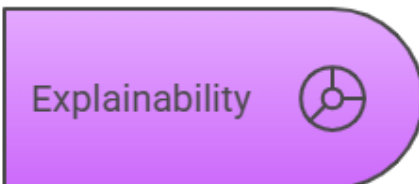
Bullseye icon representing target accuracy, aiming for $\geq 80\%$ in multimodal classification.

Bar chart icon showing evaluation metrics like precision, recall, F1-score, and ROC-AUC.



Brain icon representing validation against clinical benchmarks for score validation.

Magnifying glass icon showing explainability verified through SHAP & LIME outputs.



DEMENTIA SCREENING VIA SPEECH, FACIAL EXPRESSIONS, AND TASK BEHAVIOUR

How It Will Be Validated:

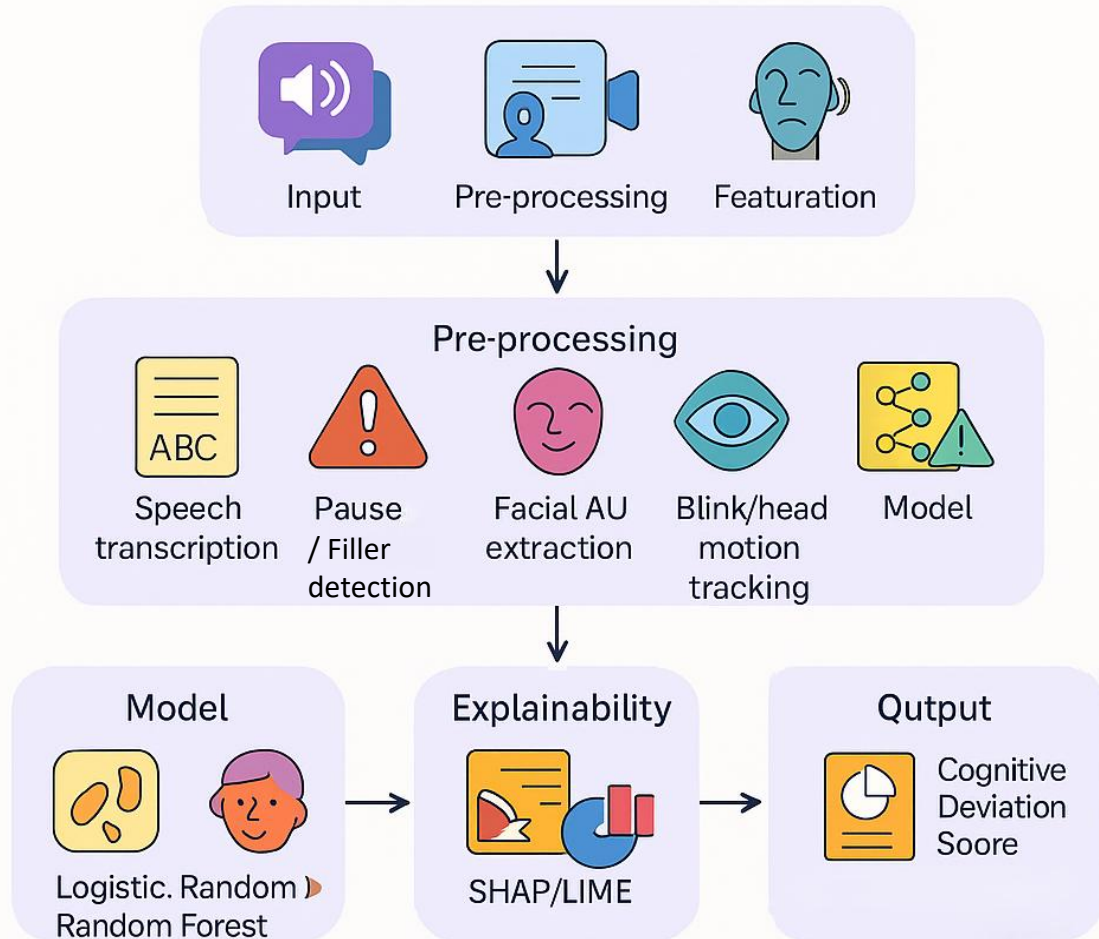
- **Train/Validation/Test splits** on DementiaBank (speech), ADRess, and PROMPT datasets.
- Cross-validation to ensure **generalization** across patient groups.
- Error analysis on **false positives/negatives** to refine feature selection and thresholds.
- Compare Cognitive Deviation Score with **clinical benchmarks (MMSE, MoCA)**.

Advantages for Clinicians:

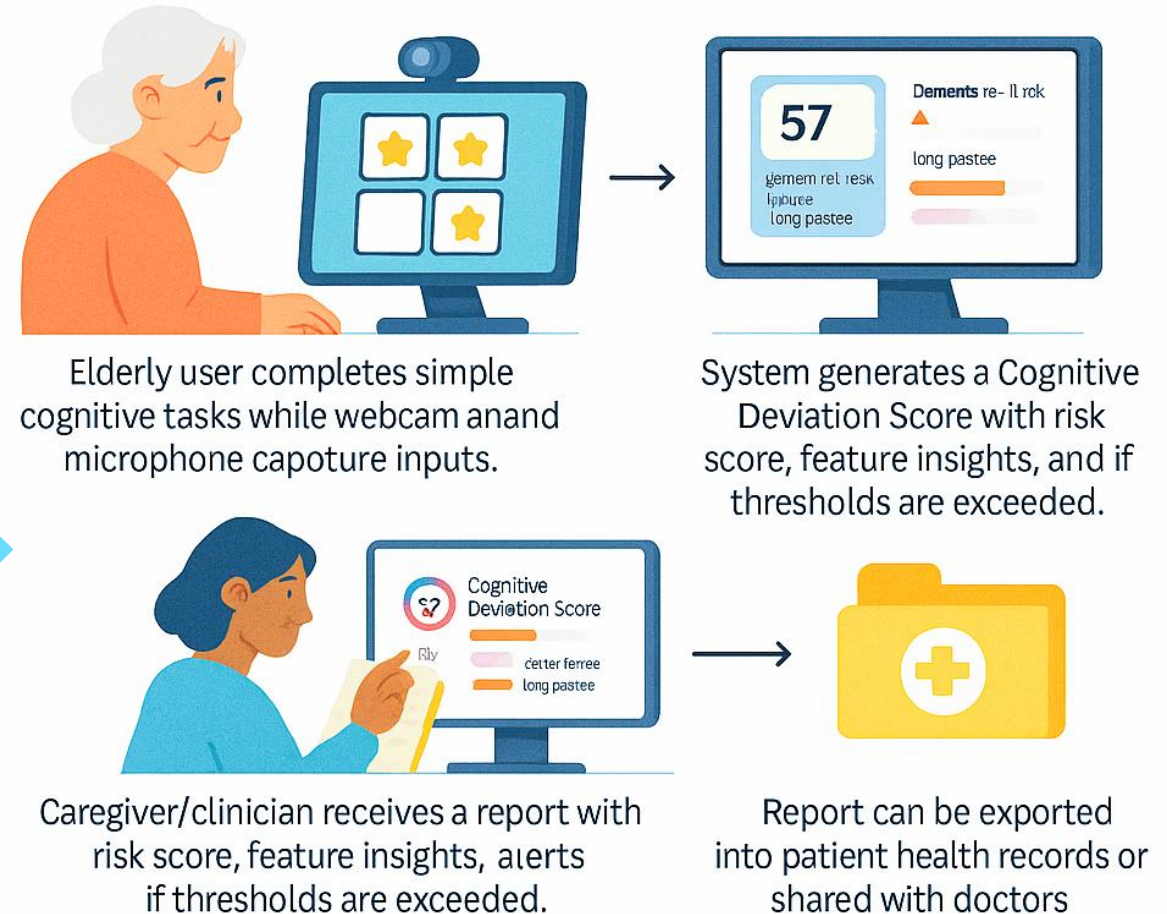
- Provides a **Cognitive Deviation Score (0–100)** with confidence levels, not just binary output.
- Explains key factors influencing results.
- Non-invasive, low-cost tool for **continuous monitoring** between clinical visits.
- Supports **early intervention** by highlighting subtle risk signs before MRI confirmation.

DEMENTIA SCREENING VIA SPEECH, FACIAL EXPRESSIONS, AND TASK BEHAVIOUR

System Overview



Real-World Usage



CogniCare: An AI-Powered Cognitive Screening & Analysis Platform

Business Strategy Framework

